

Technical Document #1012: Data Engineering - Comprehensive Overview

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Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Data lakes store raw data in its native format.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a data warehouse.
- Data governance establishes policies and procedures for managing data assets.